

DFIR Investigation Report

Case: [Insert Lab Name / Case ID]

Analyst: [Your Name/Handle]	Date: September 23, 2026
Platform: CyberDefenders	Status: Closed

1. Executive Summary

The attacker performed targeted enumeration against the HTTP service hosted on the target IIS server. Subsequently, the threat actor initiated an SMB connection and successfully planted a web-accessible payload. A misconfiguration in the SMB shares allowed anonymous/unauthorized write access, enabling the attacker to drop an ASPX webshell (`shell.aspx`). This file contained the necessary code to establish a Command and Control (C2) communication channel. Upon execution by the server, the webshell facilitated the download of the final malicious payload.

2. Methodology & Tools Used

The following forensic tools were utilized to analyze network traffic, extract memory artifacts, and perform static analysis on the identified malware:

- **Wireshark:** Employed for deep packet inspection and network traffic analysis of the provided PCAP file.
- **Volatility 3:** Utilized for memory forensics to identify rogue processes and extract injected code.
- **PEStudio:** Used for static malware analysis of the extracted executables to identify strings, imports, and capabilities.
- **VirusTotal:** Leveraged for OSINT and signature verification of extracted file hashes.

3. Incident Timeline

Note: All timestamps are recorded in UTC.

Date	Time	Event Description
2024-09-10	05:47:33	First tree connect request observed to 10.0.2.15\IPC\$
2024-09-10	05:47:33	Second tree connect request observed to 10.0.2.15\Documents
2024-09-10	05:48:52	Payload (<code>shell.aspx</code>) Create Request initiated by attacker
2024-09-10	05:48:52	Create Response generated by the server
2024-09-10	05:48:53	Malicious payload (<code>shell.aspx</code>) successfully written to the server
2024-09-10	05:49:51	Payload executed; outbound connection to C2 server established

4. Technical Findings & Artifacts

Initial Access & Execution:

The investigation revealed that malware was pushed onto the target device utilizing an open SMB connection. The Documents share was misconfigured, lacking proper access controls, which allowed the threat actor to upload the `shell.aspx` file directly into a web-facing directory.

Indicators of Compromise (IoCs):

- **Target IP Address:** 10.0.2.15
- **Attacker IP Address:** [Insert IP from PCAP]
- **C2 Domain / IP:** [Insert C2 destination from PCAP]
- **Dropped File:** `shell.aspx`
- **File Hash (SHA256):** [Insert hash you uploaded to VT]

5. MITRE ATT&CK Mapping

The observed adversary behaviors have been mapped to the MITRE ATT&CK framework to provide a standardized understanding of the attack lifecycle:

Tactic	Technique ID	Technique Name & Context
Reconnaissance	T1595	Active Scanning: Attacker enumerated the HTTP service on the target IIS server.
Initial Access	T1133	External Remote Services: Exploited misconfigured, open SMB shares to gain a foothold.
Persistence	T1505.003	Web Shell: Uploaded <code>shell.aspx</code> into a web-facing directory to execute commands.
Execution	T1059	Command and Scripting Interpreter: Server executed the ASPX payload to initiate the next stage.
Command & Control	T1071	Application Layer Protocol: Payload established outbound C2 communication.

6. Personal Reflection & Skills Development

As part of my continuous learning path in DFIR, this lab highlighted several key areas of growth that I am actively working to improve:

- **Process Identification:** I recognized a need to improve my baseline knowledge of normal Windows OS processes. I am currently studying standard process trees to better identify malicious anomalies when using Volatility.
- **IIS & ASPX Architecture:** To better understand webshell execution, I am dedicating time to learning how IIS processes ASPX files and handles backend execution.
- **Wireshark Efficiency:** Navigating the PCAP took longer than expected. I am currently building a custom Wireshark profile with specific color-coding rules for SMB and HTTP traffic to accelerate future triage.